



MRC
Biostatistics
Unit



UNIVERSITY OF
CAMBRIDGE

Adaptive Methods in Clinical Research

Lecture 6: Master protocols

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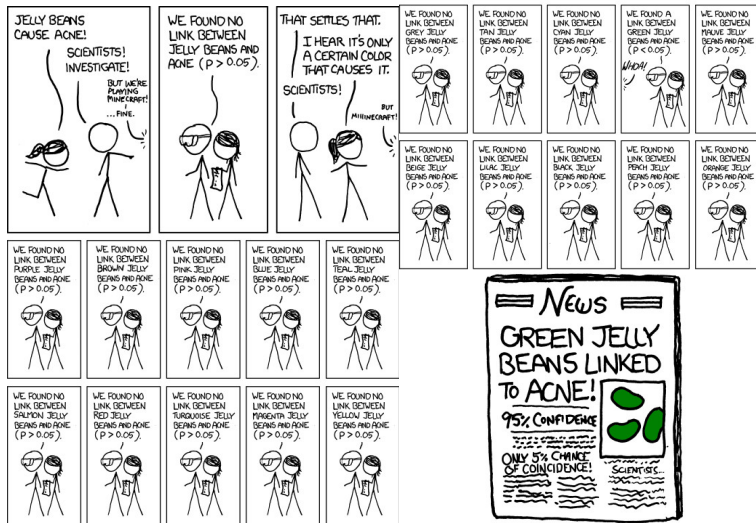
MRC Biostatistics Unit

November 2025

Master protocols are a *new* type of study that seeks to answer multiple questions within a single study

- Platform trials
- Basket trials
- Umbrella trials

Multiple testing



<https://xkcd.com/882/>

Family-wise error rate (FWER)

Weak control

$$\mathbb{P}_{\theta}(\text{Reject at least one true null hypotheses} | \mathcal{H}_0) \leq \nu.$$

Strong control

$$\max \mathbb{P}_{\theta}(\text{Reject at least one true null hypotheses}) \leq \nu.$$

False discovery rate (FDR)

Controls proportion of false discoveries among the discoveries

When to adjust for multiplicity

Parker & Weir (2020)

- Other treatment arms irrelevant for making decision about a particular treatment
⇒ Do not adjust
- When testing different subgroups, doses, ...
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- Multiple hypotheses are often tested for a single submission (e.g. Basket trial)
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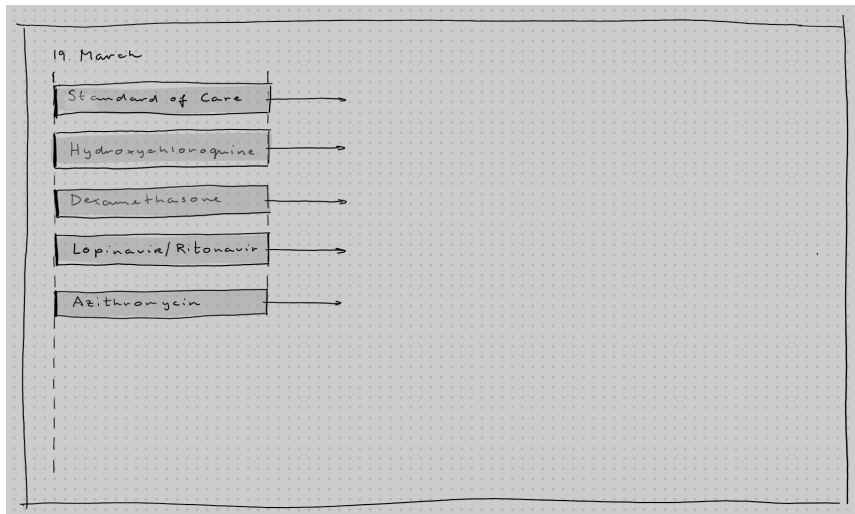
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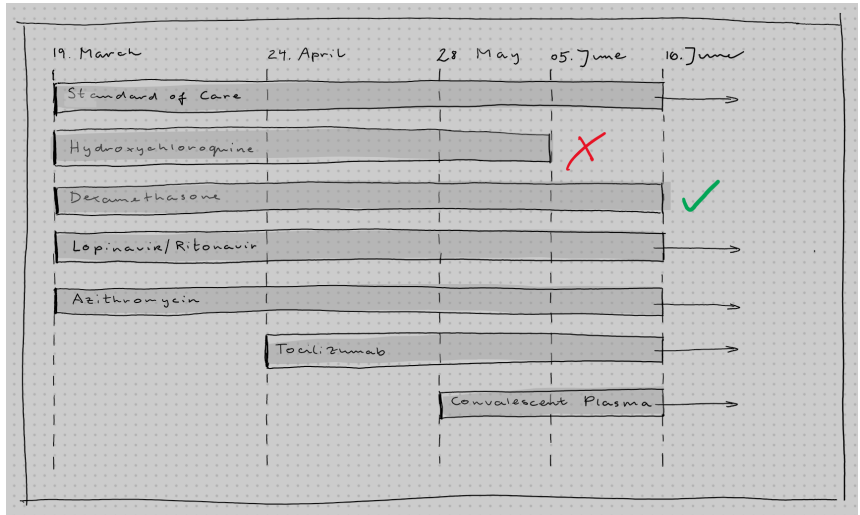
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Need to understand risk of false decisions

A platform trial (RECOVERY)



RECOVERY



What you probably didn't know

- Initial design of RECOVERY set to control FWER using methods discussed in previous lecture
- At the time not clear how to do so when adding arms

Sampling scheme (2 stage)

In stage 1 we recruit patients to all of the treatments



We then conduct an interim analysis to choose which treatments to continue recruiting from



At interim analysis j

- if $Z_{k,j} < l_j$: treatment k is dropped from trial.
- if $Z_{k,j} > u_j$: can reject H_k and stop trial.

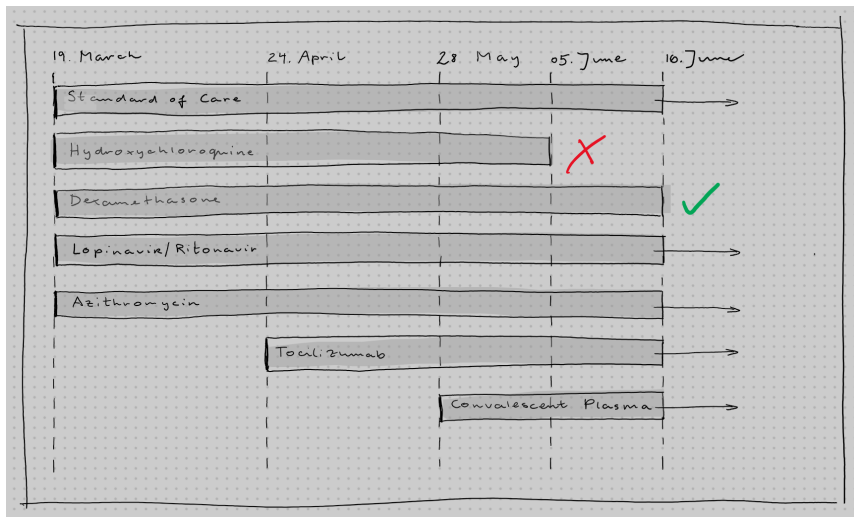
Issue: J -stage trial with K arms $\Rightarrow$ up to $K * J$ tests.

Fact: Strong control of FWER $\Leftrightarrow$ Weak control of FWER.

Problem: Test statistics are correlated.

Solution: Condition on the vector of sample means on control.

Multi-arm platform trials

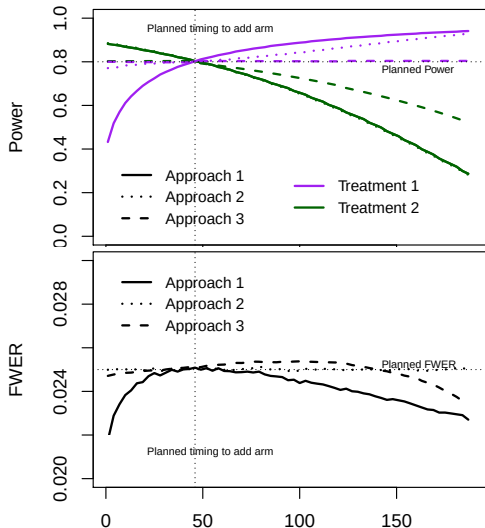


- Timing of adding an arm is known
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⇒ Same ideas as before can be used to find design

Incorrect timing



A1: Move interim to when treatment is added but keep boundaries

A2: Move interim to when treatment is added and update boundaries

A3: Keep interim

When the second active treatment is added

Advantages of pre-planned adaptive tests

1. Sufficient statistics
2. Sample size calculations
3. Confidence intervals always consistent with test decision

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BUT arm adding and treatment selection is typically more complex than a pre-planned rule.

- The conditional error, $A(Y)$, is the maximal probability of rejecting $\mathcal{H}$ with the original test, conditional on the interim data Y
- $B(Y)$ is the conditional error for a new test following an unplanned adaption.
- If $B(Y) \leq A(Y)$ the new test controls the FWER.
- Can be used to account for ad-hoc design changes

Initial design:

- $J = 3, K = 2$
- $n = 10$ patients per arm per stage
- Triangular bounds

Modification:

- At first interim
- add 2 arms

Operating characteristics

	θ	$P(\mathbf{R})$	$E_{\theta}(N)$	$P_{\theta}(R_1)$	$P_{\theta}(R_2)$	$P_{\theta}(R_3)$	$P_{\theta}(R_4)$
2 trials	(0, 0, 0, 0)	0.08	87	0.03	0.03	0.03	0.03
	(δ , 0, 0, 0)	0.98	87	0.93	0.02	0.03	0.03
	(δ , 0, δ , 0)	0.98	86	0.93	0.02	0.82	0.04
	(δ , δ , δ , δ)	0.98	84	0.81	0.81	0.77	0.77
Start new	(0, 0, 0, 0)	0.05	62 (+30)	0.02	0.02	0.02	0.02
	(δ , 0, 0, 0)	0.76	63 (+30)	0.75	0.01	0.01	0.01
	(δ , 0, δ , 0)	0.89	62 (+30)	0.67	0.02	0.67	0.02
	(δ , δ , δ , δ)	0.95	63 (+30)	0.64	0.64	0.64	0.64
Cond Error	(0, 0, 0, 0)	0.05	78	0.03	0.02	0.01	0.01
	(δ , 0, 0, 0)	0.92	71	0.91	0.00	0.01	0.01
	(δ , 0, δ , 0)	0.95	71	0.86	0.01	0.64	0.03
	(δ , δ , δ , δ)	0.98	76	0.77	0.77	0.61	0.61

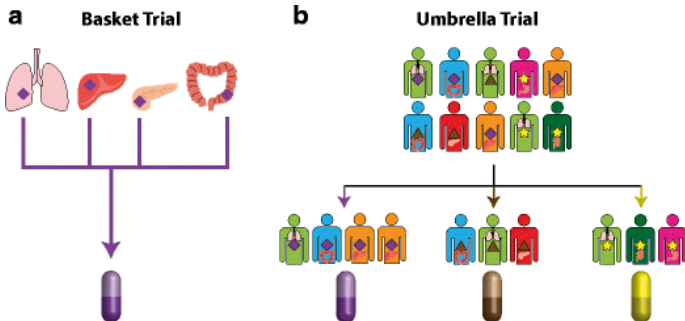
Table: R_i is the event that we reject H_{0i} , $\mathbf{R}$ is the event to reject at least one H_{0i} , $E_{\theta}(N)$ is the expected sample size.

- FWER is only one way to understand risk of wrong decisions
- Online FDR been explored in Robertson et al. (2023)
- Specific metrics have been suggested (e.g. Cui et al., 2023)

Umbrella and Basket trials

Basket Trials Single treatment and single biomarker, different histologies placed in baskets

Umbrella Trials Single histology, multiple biomarkers each matched to treatments

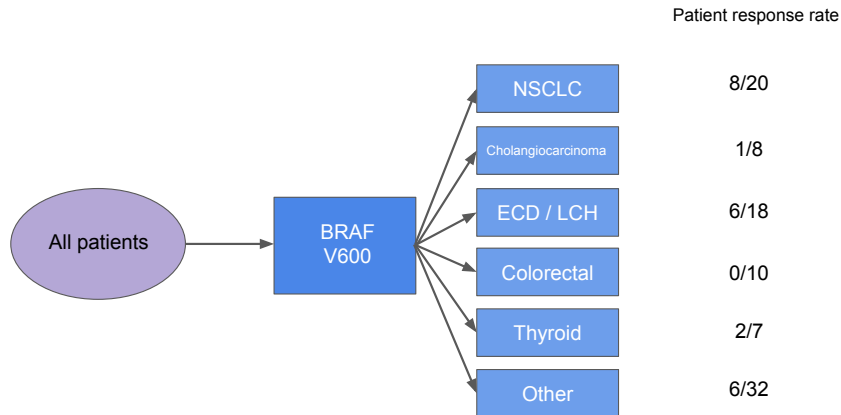


From Bui and Kummar (2018)

Basket trials in oncology – An example

Hyman *et al.* (2015) reported a recent basket trial, which has been designed to evaluate the efficacy of vemurafenib in BRAF-V600.

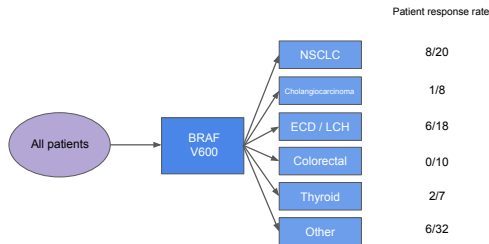
A total of 122 patients with BRAF-V600 mutations were enrolled, of which 95 entered the 6 modules.



Borrowing of information between modules

With the **common genomic mutation** targeted by the investigational drug, one may expect:

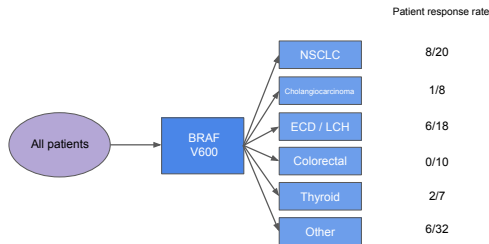
... some patient subgroups may **respond to the treatment similarly**.



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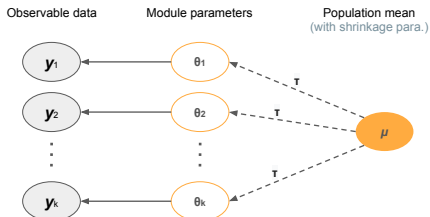
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Potential analysis strategies:

- Stand-alone analyses
- Complete pooling
- Borrowing of information

Bayesian hierarchical models



A hierarchical model for binomial data:

$$y_i | p_i, n_i \sim \text{Binomial}(p_i, n_i)$$

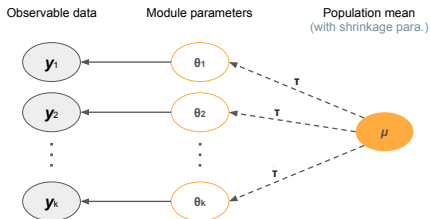
$$\text{logit}(p_i) = \theta_i$$

$$\theta_i | \mu, \tau \sim N(\mu, \tau^2)$$

$$i = 1, \dots, k.$$

- ★ Hierarchical modelling assumes **exchangeability** (similarity) of the θ_i s
- ★ The degree of borrowing is determined by τ :
 - $\tau = 0 \rightarrow$ complete pooling of data from other modules;
 - $\tau = \infty \rightarrow$ no borrowing.
- ★ Very restrictive to suppose all θ_i s will be shrunk towards one population mean

Relaxing the exchangeability assumption – EXNEX



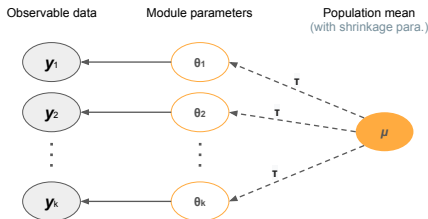
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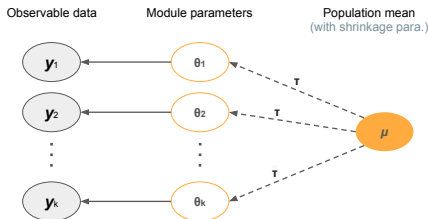
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... with probability of 1!

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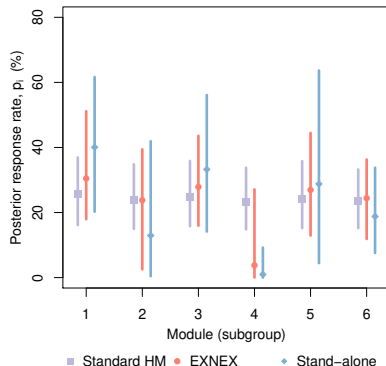
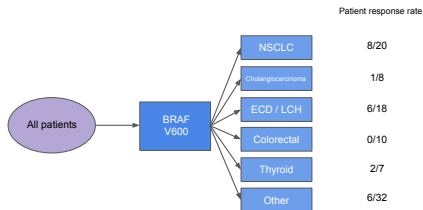
$$i = 1, \dots, k.$$

... with probability of 1!

★ Neuenschwander *et al.* (2016) propose an extension to allow for **non-exchangeability**

- ▶ **EX**: $\theta_i | \mu, \tau \sim N(\mu, \tau^2)$ with probability of w_i
- ▶ **NEX**: $\theta_i \sim N(m_i, s_i^2)$ with probability of $1 - w_i$

Comparing different analysis models



- In adaptive platform trials understanding risk of incorrect decisions is key
- Information borrowing a key feature of Basket and Umbrella trials
 - ▶ Should consistently be incorporated from design (Zheng et al, 2023) to analysis e.g. Zheng & Wason (2022)
- No one-size fits all solution for Master protocols, but case by case considerations necessary

Literature (1)

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Literature (2)

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